University of Information Technology & Sciences (UITS)
Faculty of Science and Engineering
Department of Computer Science and Engineering
Program: B.Sc. in CSE
Term Final Examination, Autumn 2023
Course Title: Electronic Devices and Circuits
Course Code: EEE 251

Marks: 50 **Time: 3 (three) Hours**

(Answer all questions)

1. a) For the Zener diode network of figure 1, determine V_R, I_L, V_Z **[05]**

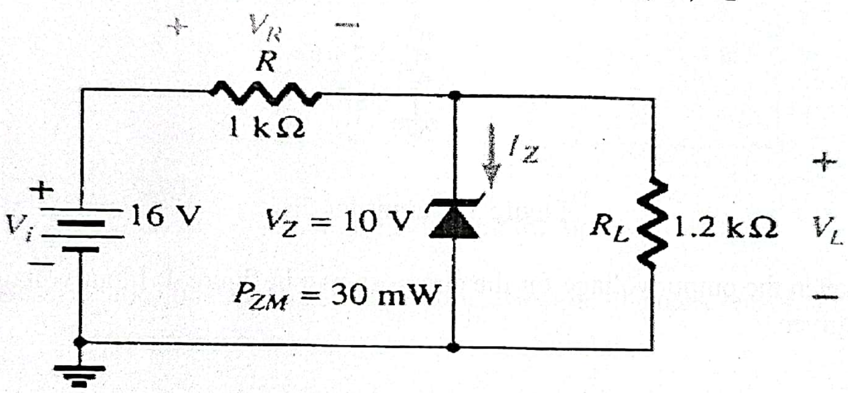

Figure 1: Circuit for 1a

b) Describe the construction and the principle of a pnp transistor. **[05]**

2. a) Given the load line of figure 2 and the defined Q -point, determine the required values of V_{CC}, R_C, and R_B for a fixed-bias configuration. **[05]**

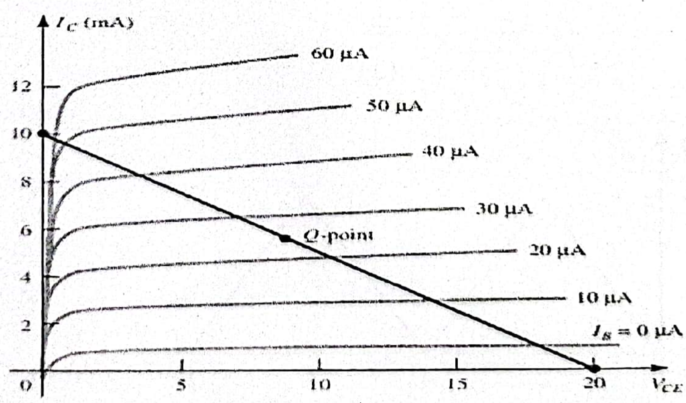

Figure 2: Circuit for 2a

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b) For the emitter-bias network of figure 3, determine I_B, I_C, V_{CE}, V_C, V_E, V_B and V_{BC}. [05]

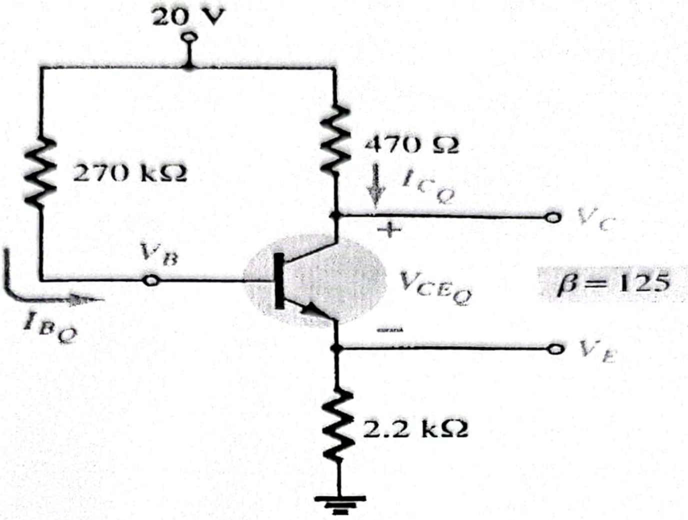

Figure 3: Circuit for 2b

3. a) Sketch the output voltage for the circuit shown in figure 4. Input voltage is given. [05]

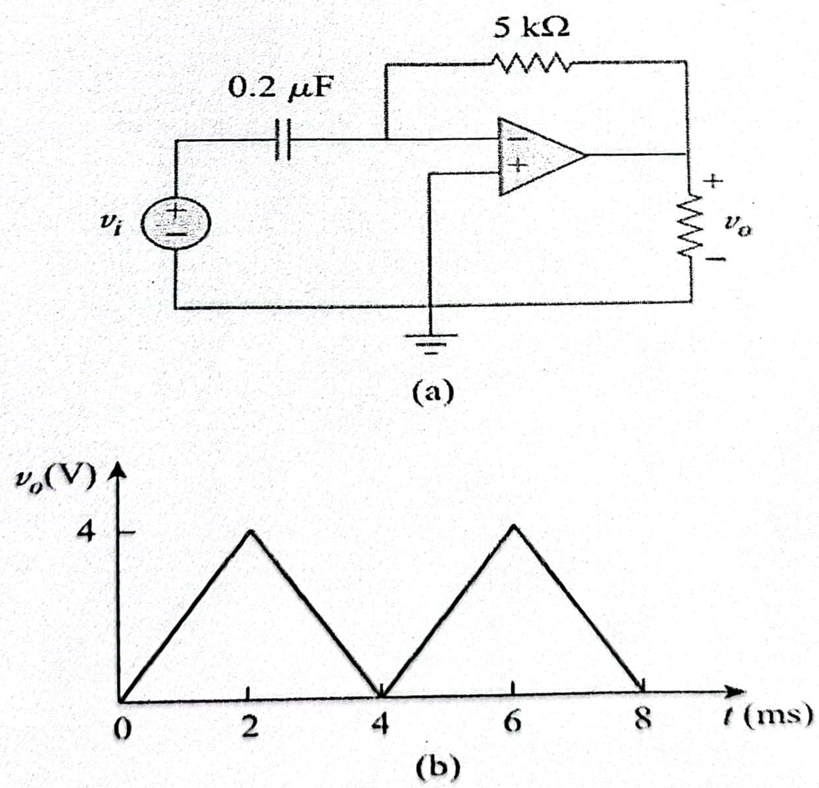

Figure 4: Circuit for 3a

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b) Explain the operation of a CMOS inverter with necessary circuit diagram. **[05]**

4. a) Explain the construction and characteristics of JFET. **[05]**

b) Explain the basic construction and characteristics of Enhancement type MOSFET. **[05]**

5. a) Determine V_O and i_O in the op amp circuit shown in figure 5 **[05]**

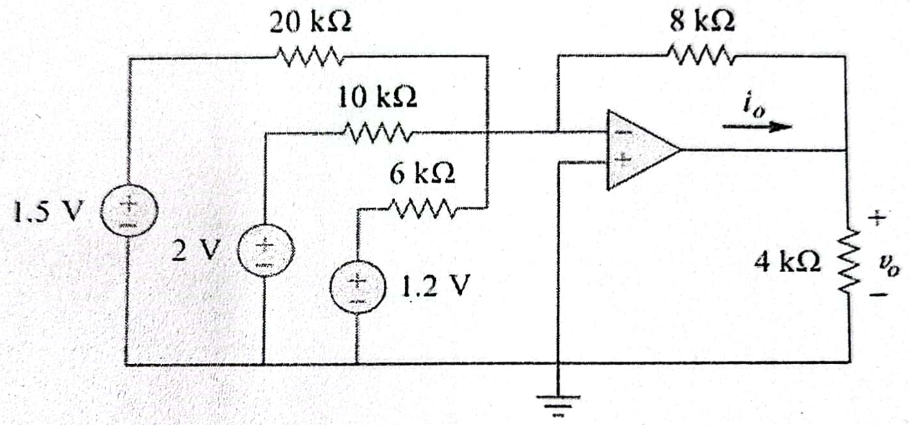

Figure 5: Circuit for 5a

b) Describe the construction and operating principle of a dark sensor circuit. **[05]**

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